Evaluating Uncertainty Label Resolution Methods for Multi-Label Chest X-Ray Classification

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I. Introduction

A. Importance

Chest radiography is one of the most widely used diagnostic imaging modalities globally, serving as a critical first line tool for screening and diagnosing life-threatening thoracic conditions [1], [2], [9]–[12]. Automated interpretation of chest X-rays (CXRs) at the level of practicing radiologists could substantially improve clinical workflow prioritization and extend diagnostic capacity in under-resourced settings [2], [3], [9], [11], [12]. Because a single radiograph can exhibit several co-occurring pathologies, automated CXR interpretation is naturally framed as a multi-label classification problem [1], [3], [9], [10], [13].

B. Background

Progress in multi-label CXR classification has been driven by large public datasets such as ChestX-ray8/ChestX-ray14 [3], [12] and CheXpert [1], [10]. More recently, resources such as CheXpert Plus [4] have expanded on these by providing paired radiology reports, patient demographics, and DICOM-format images. Because manual annotation at this scale is prohibitively expensive, image labels are instead generated automatically by natural language processing (NLP) tools that parse free-text radiology reports [1], [3], [12], [13], [15]. However, radiology reports frequently contain hedging language [1] that reflects genuine diagnostic ambiguity [12], [15] rather than a definite finding. To capture this, the CheXpert labeler assigns each of 14 observations one of four values: positive, negative, uncertain, or not mentioned [1]. Depending on the pathology, uncertain labels can constitute a substantial share of the dataset; for example, the uncertain label for Consolidation is nearly twice as prevalent (12.78%) as the positive label (6.78%) [1].

C. Problem Statement

There is currently no consensus method for resolving these uncertain labels during model training [1], [5]–[8], [10], [11], [15]. Irvin et al. [1] proposed five candidate heuristics (U-Ignore, U-Zeros, U-Ones, U-MultiClass, and U-SelfTrained) and found that the best-performing policy is pathology dependent rather than universal. Subsequent work has proposed alternative resolutions: replacing discrete labels with continuous probabilities via label smoothing [5], relabeling reports with a vision-supervised language model [6], resolving missing/uncertain labels through semi-supervised pseudo-labeling [7], and, in principle, quantifying uncertainty explicitly via Bayesian inference [8], [11]. Parallel work improves multi-label CXR classification without resolving report-level uncertain labels. Architecture and representation studies include regression into an interpretable two-dimensional label-embedding space [9], graph-refined label embeddings with ViT–DenseNet feature fusion [10], hierarchical attention with a feature-decoupling loss [13], and ConvNeXt features paired with BioBERT-derived label representations [16]; [9], [13], and [16] do not model uncertain labels, and [10] adopts U-Zeros as a default mapping. Image-based confidence estimation studies include fuzzy inference over Monte Carlo dropout entropy [11] and subjective-logic uncertainty that is higher for cases whose labels were later corrected by radiologist consensus [12]. These derive confidence from the image rather than from the clinician's hedging language, and [11] still applies U-Ones to CheXpert's uncertain labels, so they complement rather than resolve the uncertain label. Noisy-label learning [14], [15], [17] offers related tools but mostly targets incorrect or synthetically corrupted labels rather than these explicit uncertainty categories.

D. Research Gap

Three critical gaps exist in the current literature. First, uncertainty resolution strategies (heuristic mapping, label smoothing) and uncertainty quantification strategies (Bayesian methods) are rarely evaluated together on the same model and data split [5], [8], [11], [12], [15]. Second, evaluations are frequently confounded by varying architectures (e.g., DenseNet vs. ResNet) [9], [10], [12], [13], [16] and different data subsets and internal validation splits [7], [9]–[13], as well as by differences in pretraining and noise-handling strategy [14], [15], making direct cross-paper comparisons unreliable. Third, while heuristic mapping is a common baseline [1], [10], [11], it treats uncertainty as an ad-hoc pre-processing choice rather than modeling the true degree of clinical doubt [5], [7], [8], [11], [12].

E. Evidence

The practical consequences of this gap are visible in the range of results reported across studies that cannot be attributed cleanly to any one factor [1], [5], [6], [10], [13], [16]. Irvin et al. [1] reported test AUCs of 0.97 for Pleural Effusion and 0.85 for Atelectasis using heuristic label mapping on the CheXpert test set. Pham et al. [5], using conditional training combined with label smoothing across a six-model ensemble, reported a mean validation AUC of 0.940 and an independent test AUC of 0.930. Jain et al. [6] showed that improving label quality directly, without changing the downstream classifier, raised the

average F1 score from 0.54 (a naive heuristic baseline) to 0.68 using their VisualCheXbert relabeling approach. Architecture-focused studies show a similar spread on ChestX-ray14: Yang and Wu [13] report AUC values above 0.94 for all pathologies under a 70%/30% split, whereas Shi et al. [10] and Jin et al. [16] report mean AUCs of 0.849 (official protocol) and 0.826, respectively. On CheXpert, Shi et al. [10] report a mean AUC of 0.815 under an internal 70%/10%/20% split with U-Zeros as the default uncertain-label mapping. Noisy-label studies likewise target incorrect or synthetically corrupted labels [14], [15], for example jointly addressing label noise and class imbalance [17], rather than the explicit uncertainty categories in CheXpert. Because these gains arise from different splits, architectures, and label policies, it remains unclear how much of the reported improvement is attributable to the uncertainty resolution strategy itself versus other confounds such as ensembling [5] or model capacity [10], [14], [16].

F. Aims

This project aims to conduct a systematic, controlled evaluation of uncertainty label resolution methods for multi-label chest X-ray classification using the CheXpert dataset [1].

G. Objectives

1) Review and categorize existing approaches to handling uncertain labels in multi-label CXR classification [1]–[17].
2) Implement a common evaluation protocol (fixed split, convolutional backbone, and metric set) using the CheXpert dataset [1].
3) Compare heuristic label mapping strategies (U-Ignore, U-Zeros, U-Ones, U-MultiClass) [1], [10], [11].
4) Evaluate label smoothing [5] and Bayesian uncertainty-quantification [8], [11] strategies within the same controlled protocol.
5) Identify which resolution strategy generalizes best across pathologies with differing uncertain label prevalence [1].

II. Literature Review

A. Critical Discussion

The reviewed literature falls into six broad methodological groups regarding how clinical uncertainty, label noise and multi-label supervision are handled.

Heuristic label mapping: Early datasets used NLP to extract disease mentions but forced them into binary categories, introducing label noise [3]. To address this, Irvin et al. [1] proposed five heuristic mapping policies on CheXpert. Their experiments showed that no single heuristic dominates across all pathologies: U-Ones outperformed U-Zeros on Atelectasis, while U-MultiClass was superior for Cardiomegaly [1]. Rajpurkar et al. [2] and Wang et al. [3] similarly relied on hard, rule-derived labels without a distinct uncertainty category, which is effective for large-scale training but discards information about genuine diagnostic ambiguity present in the source reports.

Supervised learning with label smoothing: Recognizing that forcing binary values onto inherently ambiguous findings is unnatural, Pham et al. [5] replaced discrete uncertain labels with continuous probability targets via label smoothing regularization, combined with a hierarchy-aware conditional training procedure, achieving a mean test AUC of 0.930. However, because this result depends on a six-model ensemble with test-time augmentation, isolating the specific contribution of label smoothing from the effect of ensembling is difficult [5].

Vision-language relabeling: Rather than altering the training loss, Jain et al. [6] targeted label quality directly: using a vision-supervised BERT model (VisualCheXbert) to relabel report text raised the downstream average F1 score from 0.54 to 0.68. Chambon et al. [4] complemented this direction by releasing CheXpert Plus, which pairs de-identified free-text reports, DICOM metadata, and demographic data with the original CheXpert images, improving the quality and richness of the underlying data without prescribing how a vision classifier should handle uncertainty during backpropagation.

Semi-supervised and Bayesian uncertainty quantification: A more recent line of work treats uncertain and missing labels as an inference problem rather than a fixed pre-processing choice. Ihler et al. [7] adapted FixMatch to the multi-label, class-imbalanced setting of CheXpert, using distribution alignment to generate reliable pseudo-labels for uncertain and unmentioned observations, improving AUC from 0.7794 to 0.8004 on a 5k-image subset. Yang et al. [8] instead used Monte-Carlo Dropout to produce explicit predictive uncertainty estimates, reporting higher predictive entropy on clinically ambiguous cases. Ghesu et al. [12] treat uncertainty as a learned model output using subjective logic, and their rejection experiments raise average ROC AUC from 0.85 to 0.93 once the most uncertain cases are withheld. Dahan et al. [11] extend the Monte Carlo dropout idea by feeding dropout entropy into an adaptive fuzzy inference layer, reducing expected calibration error (ECE) by about 41% under cross-dataset shift. These estimators derive confidence from the image rather than from the clinician's hedging language, so they complement, rather than replace, the CheXpert-style uncertain label handled by Irvin et al. [1] and Pham et al. [5], and this line of work has not yet been benchmarked on the same AUROC-based protocol as the others.

Multi-label architectural and label-embedding methods: Rocha et al. [9] recast multi-label classification as regression into a two-dimensional label-embedding space, gaining an inherently interpretable, retrieval-based explanation and an 8.3% F1 improvement over a multi-label ResNet50 baseline; however, this is evaluated on a reduced ChestX-ray14 subset with at most three co-occurring pathologies and no uncertain category. Shi et al. [10] fuse Vision Transformer and DenseNet features through bidirectional cross-attention and initialize a Transformer decoder with graph-refined label embeddings, reaching mean AUC 0.849 on ChestX-ray14 and 0.815 on CheXpert, but apply a single U-Zeros mapping without comparing policies. Yang and Wu [13] addressed feature coupling with a hierarchical attention CNN and an orthogonal, KL-divergence-constrained feature decoupling loss that separates correlated findings such as infiltration and effusion, achieving a weighted F1 score of 0.97, Hamming loss

TABLE I. LITERATURE SURVEY OF MULTI-LABEL CHEST X-RAY CLASSIFICATION & LABEL RESOLUTION

Reference	Year	Method	Dataset	Evaluation Metrics	Performance Scores	Key Limitations
Irvin et al. [1]	2019	DenseNet-121 baseline with five heuristic uncertainty policies (U-Ignore, U-Zeros, U-Ones, U-MultiClass, U-SelfTrained)	CheXpert (224,316 CXRs; 200-study val. set; 500-study test set)	AUROC (5 competition tasks), AUPRC	Val. AUC: Atelectasis 0.858 (U-Ones), Cardiomegaly 0.854 (U-MultiClass); Test AUC: Pleural Effusion 0.97, Cardiomegaly 0.90, Atelectasis 0.85	Best performing policy varies by pathology; labels derived from text-mining rules rather than direct image review
Rajpurkar et al. [2]	2017	121-layer DenseNet (CheXNet) with weighted binary cross-entropy loss	ChestX-ray14 (112,120 CXRs; 14 pathologies)	AUROC, F1 (vs. radiologists)	F1 = 0.435 vs. radiologist average 0.387 (pneumonia); exceeded prior state of the art AUROC on all 14 classes	No explicit uncertainty labels; labels are single hard values extracted by an automatic NLP labeler
Wang et al. [3]	2017	CNN benchmarks (AlexNet, GoogLeNet, VGG-16, ResNet-50) with Class Activation Mapping–style pooling; rule-based NLP mining (DNorm + MetaMap with custom negation/uncertainty rules)	ChestX-ray8 (108,948 CXRs, 8 pathologies), later extended to ChestX-ray14 (112,120 CXRs, 14 pathologies)	NLP labeling F1 (vs. OpenI); classification AUROC; localization accuracy/AFP	Labeling F1 = 0.90 (ChestX-ray8 vs. OpenI), rising to 0.944 macro-average on the ChestX-ray14 extension; classification AUROC up to 0.8141 (Cardiomegaly, ResNet-50)	Uncertain mentions are collapsed to a binary label rather than retained as a distinct class; considerable label noise
Chambon et al. [4]	2024	Multimodal dataset curation linking de-identified free-text reports, DICOM metadata, demographics, and CheXbert derived labels	CheXpert Plus (223,228 CXR–report pairs; 64,725 patients)	De-identification recall/error rate; labeler F1 vs. image- and text-based human test sets	Only 23 of 853,878 true PHI spans fully missed by the de-identification model (0.002%), 841 partially missed (0.01%)	Does not introduce a new training-time uncertainty-handling loss; several pathologies (fracture, lung lesion, pleural other) remain severely underrepresented
Pham et al. [5]	2020	Conditional training exploiting a disease hierarchy, combined with label-smoothing regularization (LSR); six-architecture ensemble with test-time augmentation	CheXpert (223,414 training studies; 200-study val.; 500-study test)	Mean AUROC across 5 competition tasks	Validation mean AUC = 0.940 (ensemble); independent test mean AUC = 0.930, outperforming 2.6 of 3 benchmarking radiologists	Predefined disease hierarchy is imperfect and hard to extend; gains from LSR are difficult to isolate from the six-model ensemble and TTA
Jain et al. [6]	2021	VisualCheXbert: biomedically pre-trained BERT fine-tuned to predict image-derived (DenseNet) labels directly from report text	CheXpert & MIMIC-CXR (BERT trained on MIMIC-CXR reports; evaluated on CheXpert 500-study test set)	Average and weighted-average F1 (vs. radiologist image ground truth)	Average F1 rose from 0.54 (Zero-One heuristic baseline) to 0.68 (VisualCheXbert); improvement of 0.14 (95% CI 0.12–0.17) over the baseline	Trained/evaluated only on the Impression section; addresses label-extraction quality rather than the training-time loss used by the downstream image classifier
Ihler et al. [7]	2024	Distribution-Aware Multi-Label FixMatch (ML-FixMatch+DA): semi-supervised pseudo-labeling with distribution alignment for missing/uncertain labels	CheXpert5k (5,000 labeled + up to 120,000 unlabeled CXRs; 5 pathologies)	AUC, AUPRC, F1	AUC improved from 0.7794 (supervised baseline) to 0.8004 (ML-FixMatch+DA), $N = 32$; similar gains (0.7942) when only exploiting the dataset's own missing/uncertain labels	Sensitive to the pseudo-label confidence threshold; evaluated on a 5k-image subset rather than the full CheXpert training set
Yang et al. [8]	2019	Bayesian neural networks with Monte-Carlo Dropout for predictive uncertainty estimation	CheXpert & ChestX-ray14	Predictive entropy and variance.	Higher predictive entropy reported on ambiguous cases, reducing overconfident misdiagnoses	High computational inference cost; no standard multi-label AUROC benchmark scores reported.
Rocha et al. [9]	2024	Label embedding regression with retrieval based explainability	ChestX ray14 subset (up to 3 co occurring pathologies), 14 labels collapsed into 5 broad categories	Accuracy, precision, recall, F1	Best rule set gave 8.3 percent F1 gain and 12.9 percent precision gain over multilabel baseline	Reduced 14 to 5 label categories so not directly comparable to standard benchmarks, weak precision on rare classes, depends heavily on autoencoder quality

of 0.086, and AUC values above 0.94 on ChestX-ray14. Jin et al. [16] instead combined ConvNeXt image features with BioBERT-derived label representations and a dual weighted metric loss, achieving a mean AUC of 0.826 on the same dataset. These approaches improve multi-label representation and class-imbalance handling, but [9], [13], and [16] do not model uncertain labels and [10] applies a single fixed mapping, so their gains reflect architecture and representation design rather than establishing how uncertainty should be resolved during training.

Robust learning with noisy labels: Khanal et al. [14] compared Vision Transformers and CNNs under synthetically corrupted medical image labels, showing that self-supervised pretraining can substantially improve ViT robustness to label noise. More broadly, Wei et al. [15] reviewed 60 studies on noisy labels and label uncertainty in medical deep learning, identifying robust loss functions, weighting, and curriculum learning as common handling strategies. Li et al. [17] proposed a multi-stage framework using invariant risk minimization and class-aware noise detection to jointly address label noise and class imbalance. However, these methods primarily address incorrect or synthetically corrupted labels rather than the

TABLE I. LITERATURE SURVEY OF MULTI-LABEL CHEST X-RAY CLASSIFICATION & LABEL RESOLUTION (CONTINUED)

Reference	Year	Method	Dataset	Evaluation Metrics	Performance Scores	Key Limitations
Shi et al. [10]	2026	ViT plus DenseNet fusion with graph guided Transformer decoder	ChestX ray14 and CheXpert (14 label, U Zeros policy)	Per label AUC, mean AUC	Mean AUC 0.849 on ChestX ray14, 0.815 on CheXpert, exceeds SynthEnsemble and Med DGTN	High computational cost, cross dataset transfer weak without fine tuning, Grad CAM evaluation only qualitative
Dahan et al. [11]	2026	Swin Transformer with Bayesian Monte Carlo dropout and adaptive neuro fuzzy inference	ChestX ray14, CheXpert, COVID 19 Radiography, 4 broad classes	Accuracy, AUC, ECE, Brier score, predictive entropy	Accuracy 94.86%, AUC 0.975, ECE reduced 41% versus ResNet50	Only 4 broad classes not full 14 label taxonomy, fuzzy rules not clinically validated
Ghesu et al. [12]	2019	Subjective logic evidential DenseNet with uncertainty driven sample rejection and bootstrapping	ChestX ray8 and PLCO, multi radiologist relabeled test subsets	ROC AUC at varying rejection rate, F1 under bootstrapping	AUC rose from 0.85 to 0.93 at 50% rejection, effusion F1 rose from 0.60 to 0.69	Rejects uncertain cases rather than resolving them, tested on binary tasks not multi label
Yang et al. [13]	2025	Hierarchical attention CNN with orthogonal feature decoupling loss	ChestX ray14, 112,120 images, 14 labels	Weighted F1, Hamming Loss, per pathology AUC	F1 0.97, Hamming Loss 0.086, AUC above 0.94 for all pathologies, outperforms MA CNN, ML GCN, CS Net, MRChexNet	Single modality only, high model complexity, no external dataset validation, baselines reimplemented not benchmarked on shared protocol
Khanal et al. [14]	2024	ViT vs ResNet18 robustness to synthetic label noise with self-supervised pretraining	COVID-DU-Ex (33,920 CXRs) and NCT-CRC-HE-100K (100,000 training images)	Balanced accuracy (BEST, LAST), t-test	Pretraining improved ViT robustness, MAE+ViT Small significantly better at high noise levels ($p < 0.05$)	Synthetic single-label noise, no CheXpert or genuine uncertainty labels
Wei et al. [15]	2024	Scoping review of noisy label detection and handling in medical deep learning	60 studies from 2016 to 2023 across medical prediction tasks	Review based comparison of detection, handling and evaluation methods	60 studies included. robust loss, weighting and curriculum learning identified as practical approaches	Review rather than experimental study, heterogeneous datasets and methods limit direct numerical comparison
Jin et al. [16]	2023	ConvNeXt with BioBERT label embeddings and dual weighted metric loss	ChestX ray14 (112,120 CXRs, 14 pathologies)	Per label AUC, mean AUC	Mean AUC = 0.826, outperforming comparison models	No explicit uncertainty handling, evaluated only on ChestX ray14
Li et al. [17]	2023	Multi stage noisy label learning with invariant risk minimization and class aware GMM	HAM10000 and CHAOYANG imbalanced noisy medical image datasets	Classification performance against noisy label baselines, ablation studies	Achieved state of the art results on both medical datasets	Multi stage training complexity, hard noisy samples may be misidentified, no CheXpert evaluation

explicit clinical uncertainty categories present in CheXpert, limiting direct comparison with uncertainty-resolution policies.

Taken together, these six groups span a spectrum from simple pre-processing heuristics [1]–[3], through learned continuous targets [5] and improved label sources [4], [6], to semi-supervised and inference-time uncertainty modelling [7], [8], [11], [12], multi-label architectural and label-embedding methods [9], [10], [13], [16], and robust learning under noisy supervision [14], [15], [17]. The later studies demonstrate that performance under imperfect supervision can also depend strongly on model architecture, pretraining, class imbalance, and the chosen noise handling strategy. However, no existing study compares representative uncertainty resolution methods under one fixed architecture, data split, and evaluation protocol, which motivates the controlled evaluation proposed here.

III. METHODOLOGY AND EXPERIMENTAL SETUP

A. Proposed Methodology

Our study is a controlled comparison. We keep the data split, the backbone, the preprocessing, the optimizer, and the evaluation the same for every experiment. The only thing we change is how CheXpert's uncertain labels (-1) are handled during training. This way, any difference in results can be linked to the label strategy and not to something else, such as the architecture, the split, or ensembling (Section I-D).

The pipeline has five stages. First, we load the CheXpert labels, keep the five competition pathologies, and build a patient-level train/validation/test split (Section III-C). Second, we convert each uncertain label into a training target using one of six policies (Table II). Labels that are "not mentioned" are treated as negative (0) in every policy, so the policies only differ in how they treat -1. Third, we train the same DenseNet-121 under each policy. Fourth, we run inference, using a normal forward pass for the five label policies and repeated stochastic passes with dropout turned on for the Bayesian variant. Finally, we evaluate all policies on the same held-out data using discrimination, calibration, and uncertainty metrics with confidence intervals.

The six policies match Objectives 3 and 4 and are listed in Table II.

We do not include U-SelfTrained in the main comparison. It needs a second training stage, which would add another difference between the methods. We leave it for future work. The MC Dropout variant still needs some label mapping during training, so we use U-Zeros as its reference. Because of this, what it adds is a predictive entropy estimate and a change in calibration, not a new label mapping.

The final goal (Objective 5) is to find which strategy works best across pathologies with different amounts of uncertainty,

TABLE II. UNCERTAINTY RESOLUTION POLICIES COMPARED

Policy	Uncertain label u	Loss
U-Ignore	Left out of the loss (masked)	Masked BCE
U-Zeros	$u \to 0$	BCE
U-Ones	$u \to 1$	BCE
U-MultiClass	Kept as a third class	3-class cross-entropy per pathology
U-Ones + LSR	$u \to$ soft target from a fixed range, e.g. $[0.55, 0.85]$ [5]	BCE with soft targets
U-Zeros + MC Dropout	$u \to 0$; uncertainty estimated at inference [8], [11]	BCE

TABLE III. TRAINING CONFIGURATION (SAME FOR ALL POLICIES)

Setting	Value
Initialization	ImageNet-pretrained DenseNet-121, all layers fine-tuned
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$), weight decay 10^{-5}
Learning rate	10^{-4}, reduced by a factor of 0.1 (patience 1) when validation mean AUROC stops improving
Batch size	32
Max epochs	10
Early stopping	Patience of 3 epochs on validation mean AUROC; best checkpoint kept
Loss	BCE with logits (masked for U-Ignore, soft targets for LSR); cross-entropy for U-MultiClass
Class weighting	None by default; positive-class weights tested as an extra experiment
Precision	Mixed precision (FP16)
Runs	One run per policy, fixed seed (42)
MC Dropout	$p = 0.3$, $T = 20$ passes at inference
Regularization	Weight decay, augmentation, early stopping

from 3.6% for Cardiomegaly to 15.1% for Atelectasis.

B. Model Architecture

We use DenseNet-121 pretrained on ImageNet as the backbone. This is the same network used in the CheXpert baseline [1] and in CheXNet [2], so our results can be compared directly with that work. It is also small enough (about 8M parameters) that training six models is realistic on our hardware, and it is a well tested convolutional baseline for chest X-rays.

We replace the original 1000-class classifier with a new head. For the five binary-style policies, the head has five independent output units, one per pathology, each with a sigmoid activation. We use independent outputs because several findings can appear in the same image. For U-MultiClass, the head has five separate 3-way softmax outputs (15 logits in total). For the MC Dropout variant, we add a dropout layer ($p = 0.3$) before the classifier and keep it active at inference.

The overall flow is as follows. The input image is converted to a normalized 3-channel tensor and passed through the DenseNet-121 feature extractor. Global average pooling gives a 1024-dimensional vector, which goes through the optional dropout layer and the linear head to produce one probability per pathology. Only the head and the label handling change between experiments.

C. Dataset and Preprocessing

We use CheXpert [1]. Its labels come from a rule-based labeler with four possible values (positive, negative, uncertain, not mentioned), as described in the dataset description. We evaluate the five competition pathologies: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. These have published benchmarks [1], [5] and different amounts of uncertain labels. We use only frontal images (AP and PA) so that the view type does not become another factor that changes the results.

The official test labels are not public, so we build our own splits. The official training set is divided at the patient level into 90% for training and 10% for internal validation. Splitting by patient makes sure no patient appears in both parts, which avoids leakage between studies of the same person. The validation part is used for early stopping, threshold selection, and model selection, and its labels are resolved with the same policy as the training data. For testing, we use the official 200-study validation set, whose labels come from the consensus of three board-certified radiologists. We never use it for tuning. Since these labels have no uncertain class, they give a fair reference that does not depend on any of our policies. The split is generated once with a fixed seed, saved to a file, and reused for all experiments.

Images are resized to 320×320 following [1], the grayscale channel is copied to three channels, and pixel values are normalized with the ImageNet mean and standard deviation. For augmentation, applied to training images only, we use random rotation of up to 10°, random resized crop with a scale from 0.9 to 1.0, and small brightness and contrast changes. We do not use horizontal flipping, because the position of the heart and the side of a finding matter for diagnosis. Validation and test images are only resized and normalized.

D. Training Configuration

Table III shows the training setup. All settings are the same for every policy. We do not tune anything per policy, and we do not use ensembling or test-time augmentation, which avoids the problem noted for [5], where the effect of label smoothing is hard to separate from the ensemble. We will keep the code, seed, and split files under version control so the experiments can be reproduced.

E. Evaluation Metrics

All metrics are computed for each pathology and then averaged, using the radiologist-consensus test labels.

The main metric is the area under the ROC curve (AUROC). It does not depend on a threshold, it is not badly affected by class imbalance, and it is the standard metric in [1], [5], so our numbers can be compared with published work. We also report the area under the precision-recall curve (AUPRC), which is more informative for pathologies that are

rare. For a threshold-based measure we report the F1 score, as in [6]. The threshold is chosen on the internal validation set by maximizing Youden's J statistic, and never on the test set.

Label smoothing and MC Dropout can change how well the predicted probabilities match real frequencies, not just how the predictions are ranked [11]. Calibration also matters if such a model is ever used clinically. For this reason we report the Expected Calibration Error (ECE) and the Brier score. For the MC Dropout model only, we also compute the predictive entropy and compare it between studies that had uncertain training labels and studies that did not. This tests whether the Bayesian estimate reflects the doubt in the original reports [8].

We give 95% bootstrap confidence intervals (1,000 resamples) on the test set and use paired bootstrap tests to compare each policy with the U-Zeros baseline. These intervals reflect the limited size of the test set, but not the variation between training runs, since each policy is trained once. The test set also has only 200 studies, so small differences may not be statistically significant, and we will say so in the results. For Objective 5, we tabulate each policy's improvement over U-Zeros next to each pathology's uncertain prevalence. With only five pathologies, this comparison is descriptive and not a formal correlation.

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